Deployment environments

Learn to understand, design, build, test and explain AI systems responsibly.

Project: Grounded Learning Service

THE CONNECTED WEEK

One complete environment-boundary record.

Keep environment, target, namespace, symbolic authority, operation and checked outcome together.

Keep one habit: Recall → W1–W3/W5 → independent development case → manifest project → delayed return.

MAIN LESSON · SECTION 1

Recall · Retrieve before notes

> Keep this task's inputs,rule and response together.

Complete task and evidence

- Try the recall before opening notes. Then use a different colour or a labelled second line to correct your explanation.

- Without opening the old lesson, explain one key idea from Queues and asynchronous work. Give an example and a mistake that the idea helps you avoid.

- Without opening the old lesson, explain one key idea from Data retention and deletion. Give an example and a mistake that the idea helps you avoid.

- Without opening the old lesson, explain one key idea from External side effects and recovery. Give an example and a mistake that the idea helps you avoid.

- Use this reference for the original W1–W3 and W5 exercises. The separately named independent case supplies its own changed facts; use those facts for that attempt before feedback.

- Development, staging and production are distinct **environment**s with explicit data and credential boundaries. Test configuration must not silently target production.

- For the classroom mock ENV selects a separate fixture store and secret namespace. Validate allowed **environment** names. A staging run cannot use production write credentials.

Queues and asynchronous work

> **My remembered idea, example and correction**

Data retention and deletion

> **My remembered idea, example and correction**

External side effects and recovery

> **My remembered idea, example and correction**
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Lesson reference for workbook practice

MAIN LESSON · SECTION 5

W1 · Explain the current example

> Keep this task’s inputs,rule and response together.

Complete task and evidence

- Development, staging and production require explicit data and credential boundaries. A screen label is not isolation. Main staging configuration points at production-labelled storage, so block before lookup. Fictional manifest: development dev-fixtures/dev-mock/dev-reader holds D; staging stage-fixtures/stage-mock/stage-reader holds Z; production approved-school/prod-mock/prod-reader holds A.

  Each symbolic credential permits only reads in its own declared environment/store. Match environment, actual target, secret namespace and trusted credential authority; reject missing fields, unknown names and write attempts. Picture correct staging reads Z, but substituting the production credential blocks even if its data remains fictional. All records and credential IDs here are harmless fixtures.

  Passing rehearsal supports release review, not automatic promotion or complete live coverage. The code performs only local dictionary checks and reads, with unchanged sources; it implements no real IAM, network boundary, secret access or deployment.

- Explain the weekly **mechanism** in your own words. Use the worked scenario to name its **input**, decision rule and **output**.

> **My reasoning and evidence**

MAIN LESSON · SECTION 6

W2 · Trace the calculation

> Keep this task's inputs,rule and response together.

Complete task and evidence

- Development, staging and production require explicit data and credential boundaries. A screen label is not isolation. Main staging configuration points at production-labelled storage, so block before lookup. Fictional manifest: development dev-fixtures/dev-mock/dev-reader holds D; staging stage-fixtures/stage-mock/stage-reader holds Z; production approved-school/prod-mock/prod-reader holds A.

  Each symbolic credential permits only reads in its own declared environment/store. Match environment, actual target, secret namespace and trusted credential authority; reject missing fields, unknown names and write attempts. Picture correct staging reads Z, but substituting the production credential blocks even if its data remains fictional. All records and credential IDs here are harmless fixtures.

  Passing rehearsal supports release review, not automatic promotion or complete live coverage. The code performs only local dictionary checks and reads, with unchanged sources; it implements no real IAM, network boundary, secret access or deployment.

- Staging configuration points to production storage while the interface displays staging. Show the intermediate steps and give the conclusion with its scope.

My reasoning and evidence

MAIN LESSON · SECTION 7

W3 · Repair a misconception

> Keep this task's inputs,rule and response together.

Complete task and evidence

- Development, staging and production require explicit data and credential boundaries. A screen label is not isolation. Main staging configuration points at production-labelled storage, so block before lookup. Fictional manifest: development dev-fixtures/dev-mock/dev-reader holds D; staging stage-fixtures/stage-mock/stage-reader holds Z; production approved-school/prod-mock/prod-reader holds A.

  Each symbolic credential permits only reads in its own declared environment/store. Match environment, actual target, secret namespace and trusted credential authority; reject missing fields, unknown names and write attempts. Picture correct staging reads Z, but substituting the production credential blocks even if its data remains fictional. All records and credential IDs here are harmless fixtures.

  Passing rehearsal supports release review, not automatic promotion or complete live coverage. The code performs only local dictionary checks and reads, with unchanged sources; it implements no real IAM, network boundary, secret access or deployment.

- A report omits this qualification: Passing staging tests does not prove production limits or operational conditions are identical.

  Explain how omitting it could change a reader's interpretation, and write an accurate replacement.

> **My reasoning and evidence**

MAIN LESSON · SECTION 9

W5 · Explain the limitation

> Keep this task’s inputs,rule and response together.

Complete task and evidence

- Development, staging and production require explicit data and credential boundaries. A screen label is not isolation. Main staging configuration points at production-labelled storage, so block before lookup. Fictional manifest: development dev-fixtures/dev-mock/dev-reader holds D; staging stage-fixtures/stage-mock/stage-reader holds Z; production approved-school/prod-mock/prod-reader holds A.

  Each symbolic credential permits only reads in its own declared environment/store. Match environment, actual target, secret namespace and trusted credential authority; reject missing fields, unknown names and write attempts. Picture correct staging reads Z, but substituting the production credential blocks even if its data remains fictional. All records and credential IDs here are harmless fixtures.

  Passing rehearsal supports release review, not automatic promotion or complete live coverage. The code performs only local dictionary checks and reads, with unchanged sources; it implements no real IAM, network boundary, secret access or deployment.

- For the worked scenario, propose one additional case that could reveal a failure, describe its expected outcome before running it, and explain what would need to remain fixed for a fair comparison.

  Do not reuse W4 as a new final test after receiving feedback.

> **My reasoning and evidence**

MAIN LESSON · SECTION 4

Try alone · Selected transfer

> Keep this task’s inputs,rule and response together.

Complete task and evidence

- Development uses fake records but a production token. What boundary is violated?

- Attempt this case before feedback. Record any help. Earlier examples below are further practice; a case already practised with feedback cannot establish fresh transfer.

Independent case: Apply a new case independently

> **My independent explanation, trace and limits**

MAIN LESSON · SECTION 10

Project · Keep the service evidence

> Preserve the contract, inputs, observed outcomes and limits.

Complete task and evidence

- Add a labelled deployment **environment**s **evidence** sheet to Grounded Learning Service. Show how you can explain the **mechanism** and terminology of deployment **environment**s.

  Use the supplied fictional case and keep its **input**s, intermediate steps, calculation or **prediction**, and limitation. Connect this record to last week's record; a matching answer without a trace is insufficient.

My project change, supporting evidence and remaining limit

MAIN LESSON · SECTION 11

Feedback · Repair and return

> Keep this task's inputs,rule and response together.

Complete task and evidence

- Revisit this idea with your teacher after a delay. Familiarity is different from explaining and using it on a changed case.

Evidence	My record
Worked alone / hint / model	
First step I repaired and why	
New case checked later; date and result	
What I can now explain without notes	
One remaining question	

Reference bank and extra practice

Choose the reference that helps your next step.

▼ **Reference · Supported W2 record**

REFERENCE · SECTION 2

Reference · Supported W2 record

> Keep this task's inputs,rule and response together.

Complete task and evidence

- Use W2: Trace the worked case for Deployment **environment**s. Record the answer once; the lesson and workbook refer to the same attempt.

Given inputs	Procedure I chose	Middle steps	Result and check

> **The step I completed without help, and the hint I needed**

▼ Extra practice · Individual reasoning check

REFERENCE · SECTION 3

Extra practice · Individual reasoning check

> Keep this task's inputs,rule and response together.

Complete task and evidence

- Use the worked case for Deployment **environment**s. Proposed explanation: "The worked result establishes every future case". Decide whether this explanation is supported.

  Point to the step or supplied fact that decides; explain your repair if needed.

> **My explanation before partner discussion**

▼ Reference · Earlier W4 response copy

REFERENCE · SECTION 8

Reference · Earlier W4 response copy

> Keep this task's inputs,rule and response together.

Complete task and evidence

- Development uses fake records but a production token. What boundary is violated?

> **My reasoning and evidence**

▼ **Extra practice · Pictured case**

REFERENCE · SECTION 12

Extra practice · Pictured case

> Keep this task's inputs,rule and response together.

Complete task and evidence

- Separate rehearsal and live service. Staging rehearses a new answer version with fictional records; production serves approved school records.
- Staging card names trial room Z; production card names room A. Credentials and stores are separate.
- Identify the staging data and credential boundary. Why must the rehearsal stop if its credential reaches production even though the **process** is labelled staging?
- Support used: alone / hint / worked example. This record is practice **evidence**.

- Project connection: For the Grounded Learning Service, retain an annotated trace and the changed-case result. Name the contract or **evidence** that makes the deciding step valid, so another team can review it.



Follow the labelled picture with the matching explanation. Open the vector diagram

My picture-based explanation and changed-case prediction

▼ **Reference · Project milestone**

REFERENCE · SECTION 13

Reference · Project milestone

> Keep this task's inputs,rule and response together.

Complete task and evidence

- Project: Grounded Learning Service
- Current milestone: Evaluate and revise. Compare with your starting method, explain errors and record a justified revision.
- For the Grounded Learning Service, retain an annotated trace and the changed-case result. Name the contract or **evidence** that makes the deciding step valid, so another team can review it.

Evidence to keep	My record
This week's input and deciding step	
Prediction and actual result (or labelled paper trace)	
One change since the previous checkpoint and why	

Evidence to keep	My record
One error or limit and the next test	

Explain what this week adds to your project. Reuse last week's record rather than starting again.

▼ **Extra practice · Predict,check and repair**

REFERENCE · SECTION 14

Extra practice · Predict,check and repair

> Keep supported feedback separate from the independent case.

First supported question and optional hint

- Separate rehearsal and live service. Staging rehearses a new answer version with fictional records; production serves approved school records.
- Which labelled step matches this detail? A passing rehearsal supports release review, not automatic promotion.
- 1. Use staging data
- 2. Check actual connection

- 3. Review **evidence**

- Hint if needed: Read the steps in the pictorial case. Match the supplied detail to the stage that performs or records it.

Practice 1: explain the deciding evidence

> **My selected conclusion, deciding fact and explanation**
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▼ **Check after the first attempt**

After your attempt, compare with the supported explanation: Review **evidence** A passing rehearsal supports release review, not automatic promotion. An **environment** includes code, configuration, permissions and data.

A staging label alone cannot create separation.

Changed-case question and optional hint

- Changed-case practice: Point staging at the production credential. Which conclusion is supported by the supplied facts?

- 1. This one example guarantees correct results for every future **input**.

- 2. We can decide the result without examining the changed **input** or the procedure.

- 3. Stop the rehearsal because its separation boundary is breached.

- Hint if needed: Use the changed condition and the deciding step in the pictorial trace. The answer must say what follows from those facts.

Practice 2: explain the deciding evidence

My selected conclusion, deciding fact and explanation

▼ **Check after the changed attempt**

After your attempt, compare with the supported explanation: Stop the rehearsal because its separation boundary is breached. Actual access matters more than the **environment** label.

Repair and retain the record

Review Queues and asynchronous work (ai-6/22). Goal: Explain the **mechanism** and terminology of queues and asynchronous work. Short practice: Explain what the queued J12 submission response proves and what it does not.

Which event permits succeeded status, and what status is honest after a worker crash?. Return to this week and explain what you changed.

Review Data retention and deletion (ai-6/20). Goal: Explain the **mechanism** and terminology of data retention and deletion. Short practice: Apply the seven-day boundary to T1 and T2.

How many T2 locations need verification, and may deletion be called complete if its backup remains?. Return to this week and explain what you changed.

Review External side effects and recovery (ai-6/17). Goal: Explain the **mechanism** and terminology of external side effects and recovery. Short practice: Distinguish a missing printer acknowledgement from confirmed failure.

What should happen if J6 is complete, and what **evidence** changes when J6 is reliably reported absent?. Return to this week and explain what you changed.

Use Download completed work in the hosted lesson to export your answers, reflections, practice feedback and code-run **evidence**. Open the HTML copy to read or print it.

Drafts stay in this browser when storage is available; clear drafts on a shared device.

What I repaired, the evidence I kept and the next test

▼ **Earlier wording, alternatives and picture transcripts**

Use this supported practice once, in the browser or on paper. Keep the reserved independent assessment for a separate first attempt before teacher feedback.

Repair a missing building block

Keep your completed work

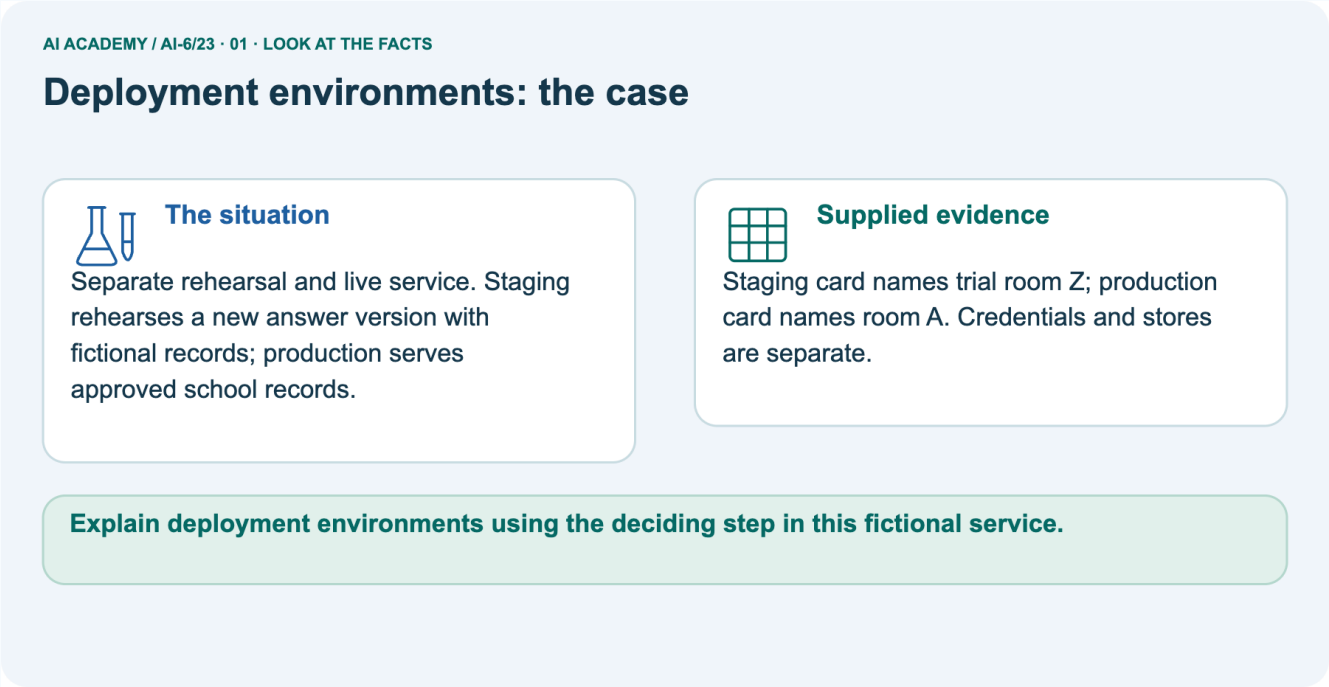
▼ **Reference · Six-picture response sheets**

REFERENCE · SECTION 15

Reference · Six-picture response sheets

> Choose the picture that helps explain the current step.

01. Deployment environments: the case



Follow the labelled picture with the matching explanation. Open the vector diagram

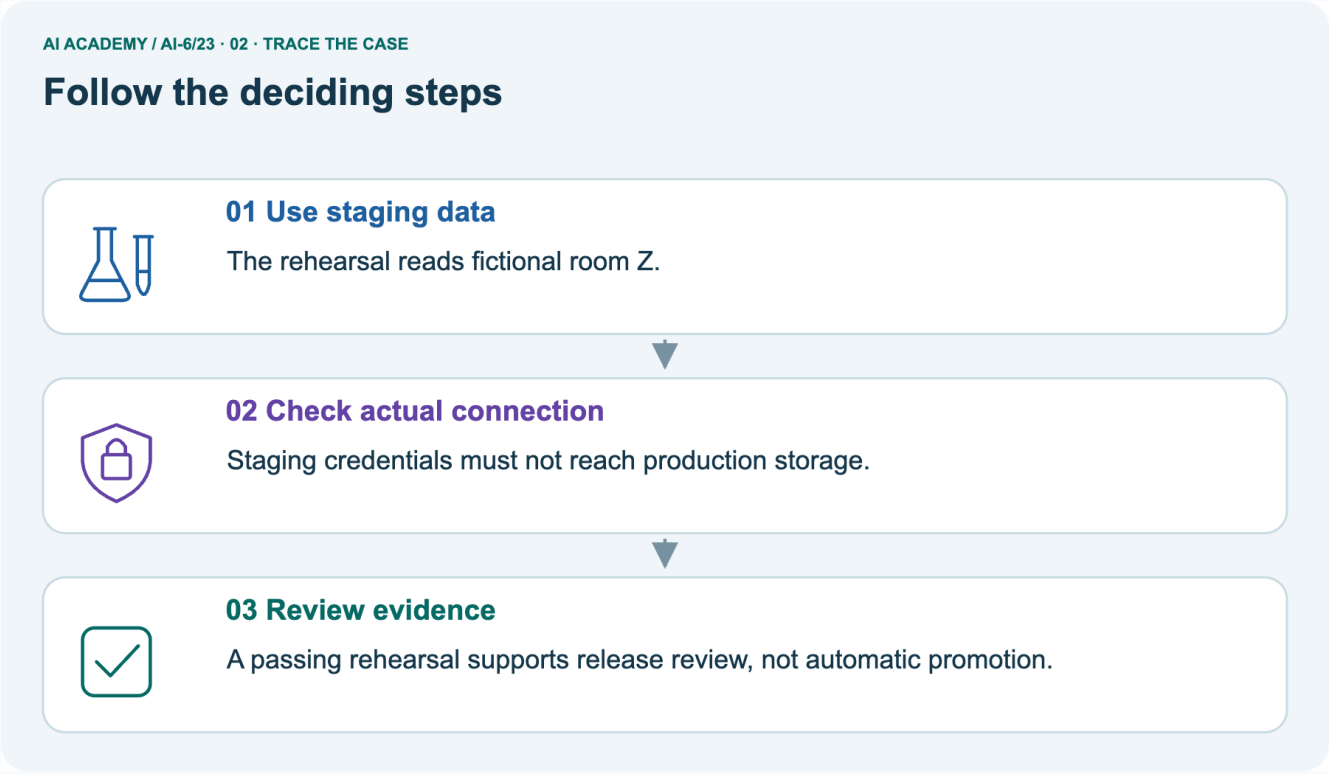
Separate rehearsal and live service. Staging rehearses a new answer version with fictional records; production serves approved school records. Supplied facts: Staging card names trial room Z; production card names room A.

Credentials and stores are separate.

Point to the given fact that matters most. Explain why it matters.

My picture explanation, deciding evidence and one limit

02. Follow the deciding steps



Follow the labelled picture with the matching explanation. Open the vector diagram

Use staging data: The rehearsal reads fictional room Z. → Check actual connection: Staging credentials must not reach production storage. → Review **evidence**: A passing rehearsal supports release review, not automatic promotion.

Point to each numbered stage and explain which recorded **input** it uses.

My picture explanation, deciding evidence and one limit

03. Deployment environments — evidence trace

AI ACADEMY / AI-6/23 · 03 · INSPECT THE LESSON MODEL

Deployment environments — evidence trace

Input and scope	Check or calculation	Conclusion and limit
Staging configuration points to production storage while the interface displays staging.	For the classroom mock ENV selects a separate fixture store and secret namespace. Validate allowed environment names. A staging run cannot use production write credentials.	The label is misleading; block the run through configuration validation. A display badge is not an isolation boundary.

The figure connects this chapter's worked input, declared procedure and result. Read the complete worked reasoning for qualifiers and intermediate steps; the abbreviated labels do not replace that reasoning.

Follow the labelled picture with the matching explanation. Open the vector diagram

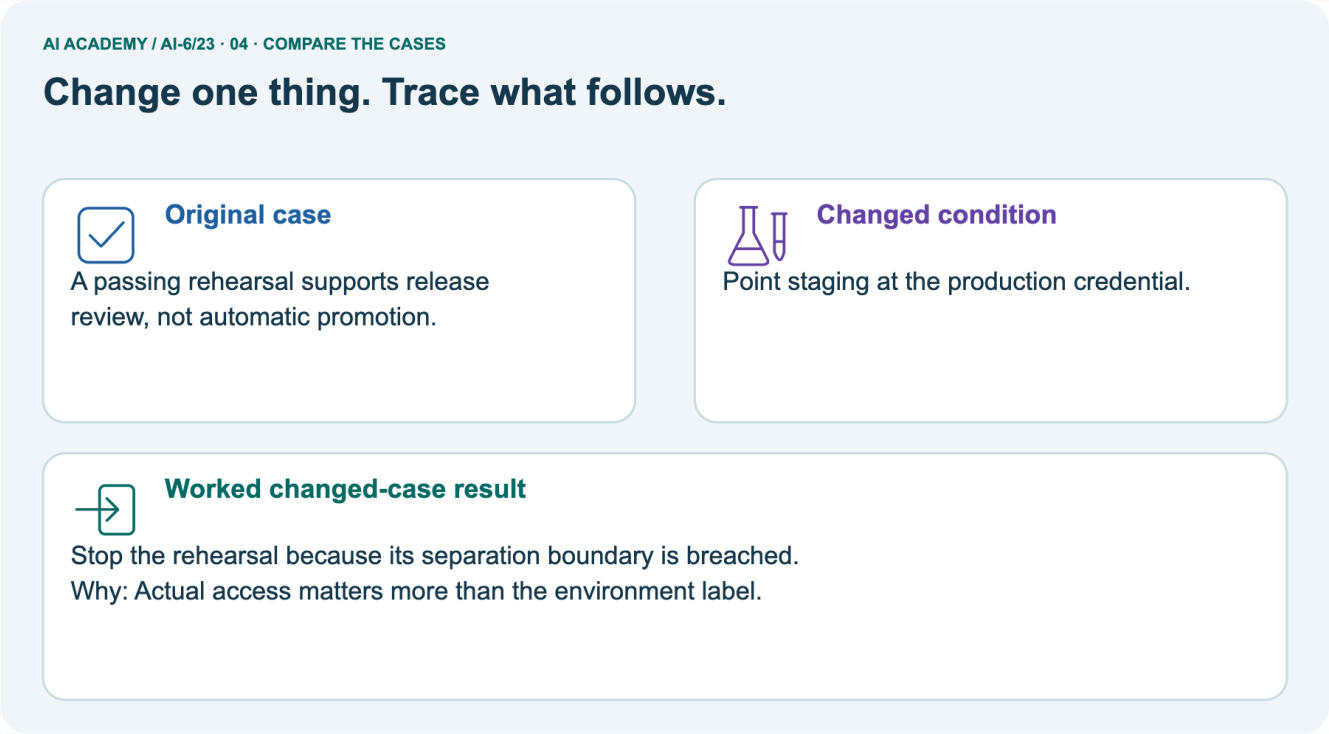
Deployment **environment**s — **evidence** trace. Staging configuration points to production storage while the interface displays staging. | For the classroom mock ENV selects a separate fixture store and secret namespace.

Validate allowed **environment** names. A staging run cannot use production write credentials. | The label is misleading; block the run through configuration validation. A display badge is not an **isolation** boundary.

Read the labels and values. Explain the relationship the model shows.

My picture explanation, deciding evidence and one limit

04. Change one thing. Trace what follows.



Follow the labelled picture with the matching explanation. Open the vector diagram

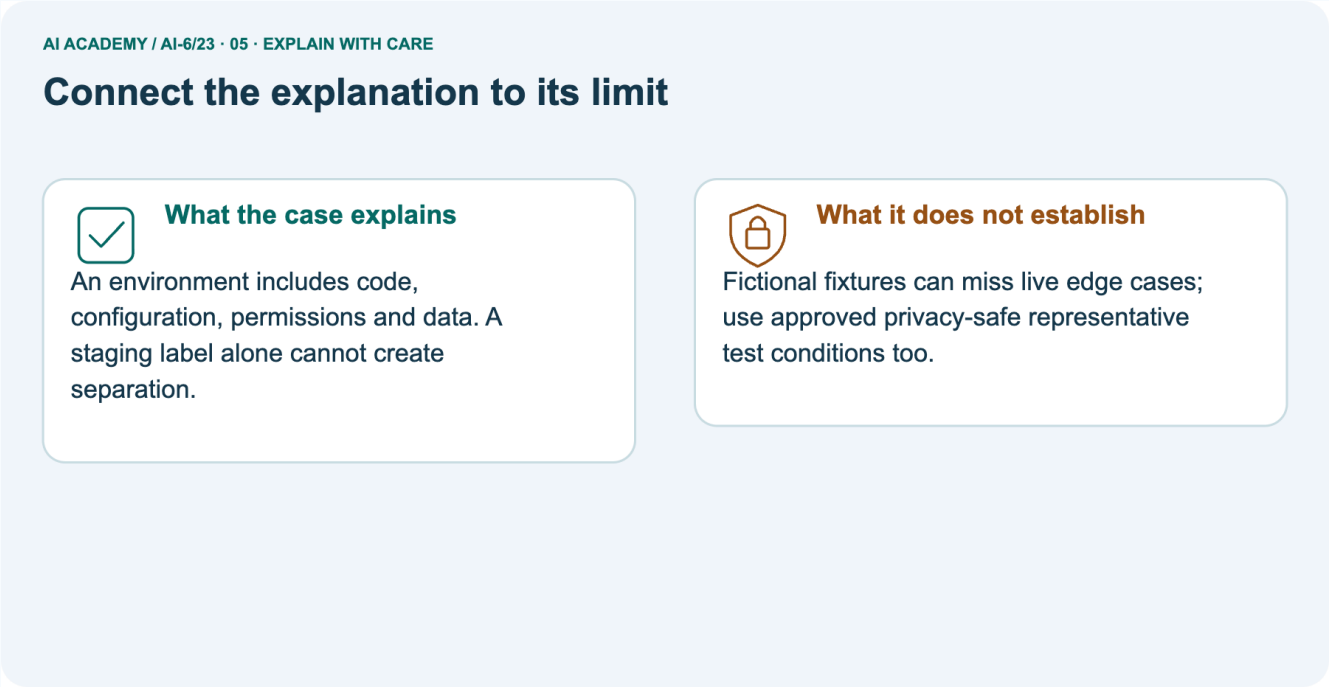
Original case: A passing rehearsal supports release review, not automatic promotion.
Changed condition: Point staging at the production credential.

Expected result: Stop the rehearsal because its separation boundary is breached. Reason: Actual access matters more than the **environment** label.

Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude.

My picture explanation, deciding evidence and one limit

05. Connect the explanation to its limit



Follow the labelled picture with the matching explanation. Open the vector diagram

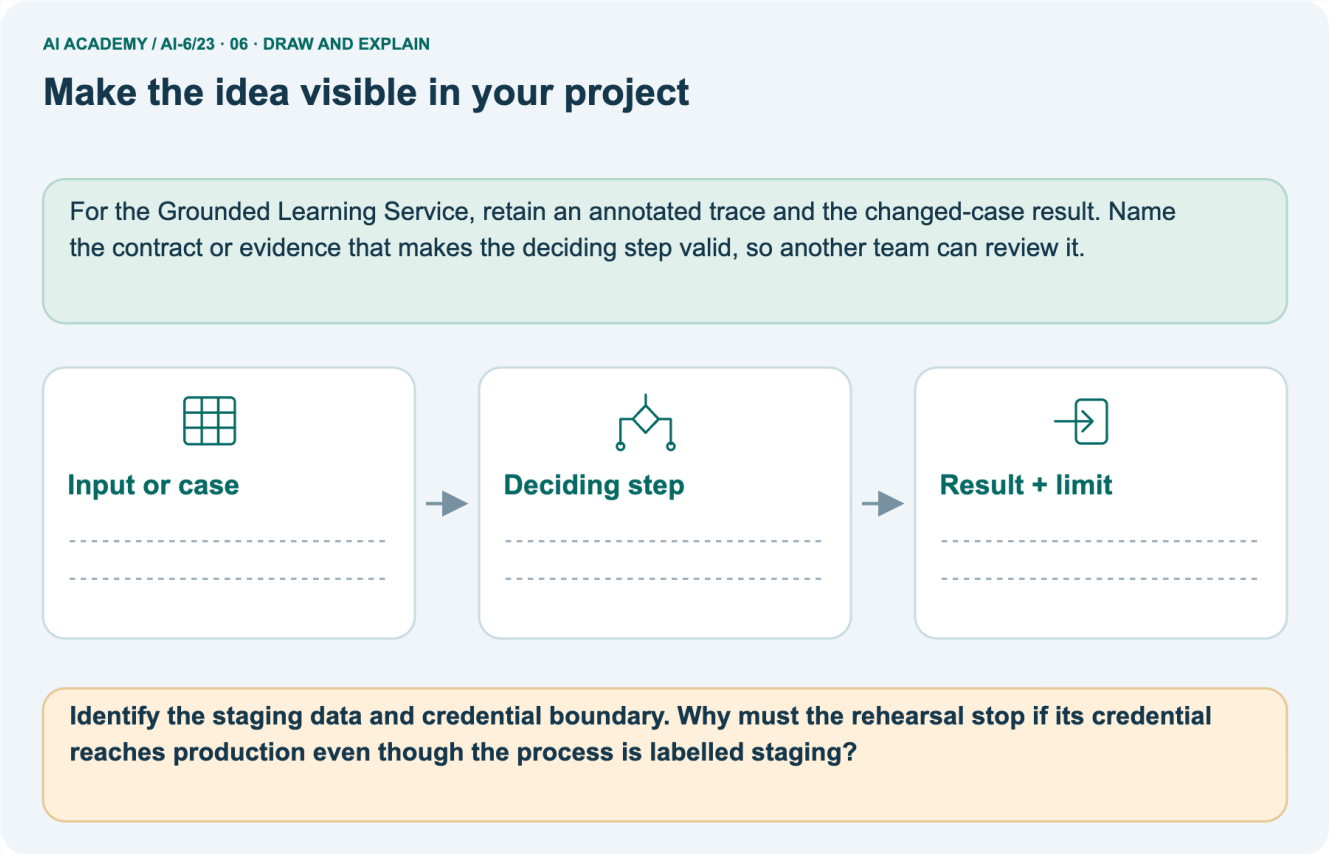
An **environment** includes code, configuration, permissions and data. A staging label alone cannot create separation.

Limit: Fictional fixtures can miss live edge cases; use approved privacy-safe representative test conditions too.

Explain the case, then point to a limit on the claim.

My picture explanation, deciding evidence and one limit

06. Make the idea visible in your project



Follow the labelled picture with the matching explanation. Open the vector diagram

For the Grounded Learning Service, retain an annotated trace and the changed-case result. Name the contract or **evidence** that makes the deciding step valid, so another team can review it.

Draw your **input**, deciding step, result and limit. Supported question: Identify the staging data and credential boundary.

Why must the rehearsal stop if its credential reaches production even though the **process** is labelled staging?

Draw your project trace and justify the deciding step using **evidence**.

My picture explanation, deciding evidence and one limit

▼ Earlier wording, alternatives and picture transcripts

01. Deployment environments: the case

02. Follow the deciding steps

03. Deployment environments — evidence trace

04. Change one thing. Trace what follows.

05. Connect the explanation to its limit

06. Make the idea visible in your project

▼ Optional interactive practice and completed work

Save or print your responses for your portfolio. Progress and drafts on the hosted website stay in the current browser. Use only fictional or teacher-approved data.